

When Not to Edit: Reliability Diagnostics for Activation-Space Interventions in Language Models

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ABSTRACT

Activation-space interventions steer language model behavior by editing hidden states along a concept direction. Existing methods usually ask how strongly to edit, but not whether the direction is reliable enough to edit at all. This is an information-reliability problem: an inferred internal signal is being used to modify generated information, and a mistaken edit can silently reduce utility, invert the target behavior, or introduce cross-concept side effects. We present a pre-flight diagnostic fingerprint for activation editing, (κ^*, d_c, A) , measuring task-subspace coherence, direction quality, and projection asymmetry. The fingerprint produces a calibrated intervention scale that can be zero, making abstention a first-class output rather than a failed edit. On GPT-2, the gated projection achieves +53.6% toxicity reduction using five-seed generation means ($p < 0.001$) while incurring $3.3\times$ less perplexity cost than ungated projection. On gender, the same framework refuses to edit because the direction is one-sided; removing the asymmetry factor over-corrects pronoun skew from 0.704 to 0.017. On Mistral-7B, the framework abstains from projection at all 32 layers ($\kappa^* \geq 40$), while an ActAdd sweep can still reduce toxicity, showing that the diagnosis is operator-specific rather than a claim that the concept is uneditable. The contribution is not a stronger universal edit, but a reliability layer that explains when projection is safe, when it should be weakened, and when it should not be used.

Index Terms—activation editing, abstention, representation engineering, concept control, safety gating, information reliability

I. INTRODUCTION

Activation-space interventions have emerged as lightweight alternatives to fine-tuning for controlling language model behavior at inference time. Activation Addition (ActAdd) [1] adds steering vectors to residual streams; Contrastive Activation Addition (CAA) [2] extends this idea with contrastive pairs; Representation Engineering (RepE) [3] extracts concept directions for behaviors such as honesty or power-seeking; Inference-Time Intervention (ITI) [4] probes for truthfulness; INLP [5] and LEACE [6] erase linearly represented concepts. These methods share a common template: estimate a concept direction $\hat{v}$ from contrastive data, then modify hidden states along that direction during generation.

The missing step is a reliability decision. A direction that is linearly decodable is not necessarily safe to edit. It may be entangled with task-relevant information, estimated from a noisy calibration split, or one-sided on neutral text so that projection pushes behavior toward the opposite pole. Existing methods normally expose a strength parameter, but they do not provide a principled procedure for returning “do not edit.” The result is a literature in which reported gains depend on layer choice, calibration seed, intervention operator, and scorer in ways that can be difficult to audit.

This paper treats activation editing as an information reliability problem. In the language of information systems, the edit is a downstream action taken on an inferred internal signal. The relevant question is therefore not only whether an edit can improve one measured outcome, but whether the signal is trustworthy enough to justify action without silently degrading other information carried by the representation. This framing connects activation editing to reliability, reject-option decision making, and safe information integration: a reliable system must know when its evidence is insufficient for action.

Our independent verification of a prior toxicity projection illustrates the problem. A previously reported GPT-2 toxicity effect at layer h.6 collapses to +1.2% under a different calibration set, while the best layer in the new run becomes h.2 at +21.6% and several layers increase toxicity. The intervention rule did not change; the calibration slice did. Blind layer selection would therefore produce an apparently reasonable but nearly inert edit.

We argue that activation editing is a decision problem (should we edit?) before it is a control problem (how much should we edit?). We introduce a three-factor pre-flight diagnostic fingerprint, (κ^*, d_c, A) , that can approve, weaken, or refuse an intervention:

- 1) **Dimension-corrected coherence** (κ^*). Measures whether the concept direction collides with the model’s task subspace after correcting for hidden-state dimension.
- 2) **Direction quality** (d_c). Measures whether the contrastive direction is statistically separable rather than a noisy artifact of finite calibration data.
- 3) **Projection asymmetry** (A). Detects one-sided directions that would move neutral text toward the opposite pole rather than remove the concept.

These factors define a calibrated scale $s =$

$s_{\max} s_{\text{gate}}(\kappa^*) s_{\text{quality}} s_{\text{asymmetry}}$, explicitly allowed to equal zero.

Our claims are intentionally limited. We do not claim the scale is optimal, that projection is always the right operator, or that a single-concept gate guarantees global safety. We claim that the fingerprint is a cheap, interpretable reliability diagnostic that predicts when projection is unsafe, explains why, and prevents the most damaging forms of blind editing. This is the paper’s novelty over our prior GBP-Audit framework [7]: the old coherence gate is adapted to high-dimensional transformer activations, expanded with quality and asymmetry diagnostics, evaluated against activation-editing baselines, and interpreted as an operator-specific reliability layer rather than as a fairness-only rule.

II. BACKGROUND AND MOTIVATION

A. Projection Editing and Its Safety Assumption

A common activation-space edit estimates a direction $\hat{\mathbf{v}}$ and modifies a hidden state $\mathbf{h}$ by subtracting the component of $\mathbf{h}$ along that direction:

$$\mathbf{h}' = \mathbf{h} - s(\mathbf{h} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}}, \quad s \in [0, 1]. \quad (1)$$

The edit is mathematically simple, but it assumes that the direction contains removable concept information rather than task-critical signal. If the concept direction is aligned with the task subspace, projection can remove the information needed to generate coherent text. If the direction is noisy, editing follows calibration noise. If neutral examples all project to the same side of the direction, projection can over-correct.

B. Ancestor Framework: GBP-Audit

GBP-Audit [7] introduced an audit-gate-act structure for tabular fairness intervention. It estimated a bias direction, measured its coherence with a task direction, and used fixed thresholds over raw cosine coherence to choose full edit, partial edit, or protection. This established abstention as a legitimate safety outcome, but three aspects do not transfer directly to transformer residual streams. First, raw cosine thresholds are dimension-dependent: a cosine that is small in tabular space can be many times larger than random expectation in $\mathbb{R}^{768}$ or $\mathbb{R}^{4096}$. Second, the old gate did not measure whether the direction was statistically reliable. Third, it could not detect one-sided directions that generate over-correction. The present paper addresses these gaps.

III. EVIDENCE OF FRAGILITY

We first document why a pre-flight reliability check is needed. This section is not a demonstration of the proposed method; it is the failure mode that motivates it.

a) Layer effectiveness is unstable.: We re-ran a toxicity projection using the standard GBP recipe on GPT-2 (124M) across all 12 layers with a calibration set drawn independently from the one used in prior work [7]. At fixed $s = 0.8$, the previously recommended h.6 layer produced only +1.2% toxicity reduction, while h.2 produced +21.6% and h.11 produced +21.2%. Other layers worsened toxicity (for example, h.8 at

−20.7% and h.10 at −23.0%). Thus the same mechanism can help, do nothing, or harm depending on layer and calibration slice.

b) Perplexity gains did not replicate.: A verification run on gender-direction projection found the largest perplexity improvement to be +0.22%, within single-run noise. We therefore treat earlier multi-percent perplexity improvements as non-replicated and do not use them as evidence for the current claim.

c) Takeaway.: Activation-editing results without abstention are conditional on calibration choices. A practitioner needs a cheap diagnostic that can reject unsafe layer-operator pairs before an expensive grid search or deployment-time failure.

IV. THREE-FACTOR RELIABILITY DIAGNOSTIC

A. Concept Direction and Task Subspace

For a concept c , let $\mathcal{P}_c$ and $\mathcal{N}_c$ be positive and negative calibration sets. At layer ℓ , we extract hidden states $\mathbf{h}_\ell^{(i)}$ and estimate a mean-difference concept direction

$$\hat{\mathbf{v}}_c^{(\ell)} = \text{normalize} \left(\frac{1}{|\mathcal{P}_c|} \sum_{i \in \mathcal{P}_c} \mathbf{h}_\ell^{(i)} - \frac{1}{|\mathcal{N}_c|} \sum_{j \in \mathcal{N}_c} \mathbf{h}_\ell^{(j)} \right). \quad (2)$$

The task subspace $\mathcal{T}_\ell^k$ is the span of the top- k right singular vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ of centered task activations. With $\mathbf{U}_k \in \mathbb{R}^{k \times d}$ containing those vectors as rows, subspace coherence is

$$\kappa_k = \|\mathbf{U}_k \hat{\mathbf{v}}\|. \quad (3)$$

All main experiments use $k = 1$ for comparability with GBP-Audit; larger- k sensitivity is discussed in Section VII.

B. Factor 1: Dimension-Corrected Coherence

In $\mathbb{R}^d$, two uniformly random unit vectors have expected absolute cosine

$$\mathbb{E}[|\cos \theta|] = \sqrt{\frac{2}{\pi d}}. \quad (4)$$

This baseline is 0.029 for $d = 768$ and 0.012 for $d = 4096$ [8]. We therefore define

$$\kappa^* = \frac{\kappa}{\sqrt{2/(\pi d)}}. \quad (5)$$

A value $\kappa^* = 1$ is chance-level alignment; $\kappa^* = 5$ is five times chance. The coherence gate is

$$s_{\text{gate}}(\kappa^*) = \sigma[-\alpha(\kappa^* - \kappa_{\text{mid}}^*)], \quad (6)$$

with $\alpha = 1.0$ and $\kappa_{\text{mid}}^* = 8.0$. The midpoint lies in the observed gap between passing and failing GPT-2 toxicity layers; the reported decisions are unchanged over a broad interval around it.

C. Factor 2: Direction Quality

A concept direction can be geometrically separable but statistically weak. We project calibration samples onto $\hat{\mathbf{v}}$ and compute Cohen’s d :

$$d_c = \frac{\bar{p}_+ - \bar{p}_-}{s_{\text{pooled}}}, \quad s_{\text{quality}} = \min \left(1, \frac{|d_c|}{d_{\text{ref}}} \right), \quad (7)$$

with $d_{\text{ref}} = 1.0$. Low d_c weakens the edit; high d_c saturates at one.

Algorithm 1 Pre-flight reliability diagnostic for activation editing

Require: Concept examples $\mathcal{P}_c, \mathcal{N}_c$; neutral prompts $\mathcal{U}$; task corpus $\mathcal{D}$; model layer ℓ .

- 1: Estimate $\hat{\mathbf{v}}_c^{(\ell)}$ from Eq. (2).
 - 2: Estimate task subspace from activations on $\mathcal{D}$.
 - 3: Compute κ^* using Eq. (5).
 - 4: Compute d_c and s_{quality} using Eq. (7).
 - 5: Compute A and $s_{\text{asymmetry}}$ using Eq. (8).
 - 6: Return $s = s_{\text{max}} s_{\text{gate}} s_{\text{quality}} s_{\text{asymmetry}}$ and diagnostic tuple (κ^*, d_c, A) .
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D. Factor 3: Projection Asymmetry

A reliable and separable direction can still be unsafe if neutral text lies almost entirely on one side of it. On a held-out neutral set $\mathcal{U}$, let $p_i = \mathbf{h}_i \cdot \hat{\mathbf{v}}$. We define

$$A = \frac{|\text{mean}(\{p_i\})|}{\text{mean}(\{|p_i|\})}, \quad s_{\text{asymmetry}} = \sqrt{\max(0, 1 - A)}. \quad (8)$$

If neutral projections are balanced around zero, $A = 0$ and no penalty is applied. If all neutral projections have the same sign, $A \rightarrow 1$ and the intervention scale goes to zero. This prevents a projection from moving neutral text toward the opposite pole.

E. Combined Scale and Algorithm

The final scale is

$$s = s_{\text{max}} s_{\text{gate}}(\kappa^*) s_{\text{quality}} s_{\text{asymmetry}}, \quad s_{\text{max}} = 0.8. \quad (9)$$

When any factor is zero, $s = 0$ and the system abstains. The same diagnostic can be used before either projection or addition, but its interpretation is operator-specific: high κ^* is especially diagnostic of unsafe projection because projection removes signal along the direction.

F. Formal Properties

Let $E_s(\mathbf{h}) = \mathbf{h} - s(\mathbf{h} \cdot \hat{\mathbf{v}})\hat{\mathbf{v}}$ be the projection edit.

Proposition 1 (Boundary consistency). *If any diagnostic factor equals zero, then $s = 0$ and $E_0(\mathbf{h}) = \mathbf{h}$ for all $\mathbf{h}$. Abstention is implemented by the same operator as editing, with no separate branch.*

Lemma 1 (Non-expansiveness). *For any $s \in [0, 1]$, $\|E_s(\mathbf{h}) - E_s(\mathbf{h}')\| \leq \|\mathbf{h} - \mathbf{h}'\|$ for all $\mathbf{h}, \mathbf{h}'$.*

Proof. $E_s(\mathbf{h}) - E_s(\mathbf{h}') = (I - s\hat{\mathbf{v}}\hat{\mathbf{v}}^\top)(\mathbf{h} - \mathbf{h}')$. The eigenvalues of $I - s\hat{\mathbf{v}}\hat{\mathbf{v}}^\top$ are 1 and $1 - s$, so its spectral norm is at most one for $s \in [0, 1]$. $\square$

These properties do not prove that every gate decision is correct; they show that refusal is a mathematically valid boundary case of the same operator. Decision quality is evaluated empirically.

TABLE I

THREE-FACTOR CALIBRATION ON GPT-2. THE SAME THRESHOLDS PRODUCE EDIT, NEAR-FULL EDIT, AND REFUSAL. ASYMMETRY VALUES OF 0.00 ARE ROUNDED; TOXICITY AND SENTIMENT HAVE RAW $A < 0.01$, WHEREAS GENDER HAS $A = 1.00$.

Concept	Layer	κ^*	d_c	s_{qual}	A	s
Toxicity	h.11	5.33	0.73	0.732	0.00	0.548
Sentiment	h.4	0.71	4.52	1.000	0.00	0.799
Gender	h.10	1.84	1.72	1.000	1.00	0.000

V. EXPERIMENTS

A. Setup

a) *Models.*: We evaluate GPT-2 (124M, $d = 768$) [9], DistilGPT-2 (82M, $d = 768$) [10], Pythia-1.4B (GPT-NeoX family, $d = 2048$) [11], and Mistral-7B ($d = 4096$) [12]. All models are frozen.

b) *Concepts and data.*: The primary concepts are toxicity, sentiment, and gender. Toxicity uses 500 toxic/non-toxic RealToxicityPrompts [13] calibration pairs and 100 high-toxicity evaluation prompts. Gender uses 100 matched pronoun-swap sentences and 20 neutral generation prompts for pronoun skew. Sentiment uses 100 positive/negative pairs. Asymmetry is evaluated on 30 concept-neutral prompts.

c) *Metrics.*: Toxicity is scored by Detoxify [14]; language utility is measured by perplexity on WikiText-103 [15]; gender skew is male/(male+female) in neutral continuations. Toxicity-effectiveness numbers in the main head-to-head table are five-seed generation means with paired-bootstrap p -values against baseline. Diagnostic calibration values are computed from the stated calibration splits.

d) *Hyperparameters.*: Unless stated otherwise, all experiments use $\alpha = 1.0$, $\kappa_{\text{mid}}^* = 8.0$, $d_{\text{ref}} = 1.0$, and $s_{\text{max}} = 0.8$. These values are fixed across models and concepts.

e) *Scope of claims.*: We report the concept-model combinations for which diagnostic values are available in the paper. We do not claim universal transfer across unreported concepts or architectures; broader replication is future work.

B. Three Outcomes on GPT-2

Table I shows that the same formula produces three qualitatively different decisions on GPT-2: a weakened edit, a near-full edit, and refusal.

Toxicity is moderately separable but noisy, so $d_c = 0.73$ pulls the scale below the ceiling. Sentiment is strong, separable, and symmetric, so the scale nearly saturates. Gender is separable and high-quality, but perfectly one-sided on neutral prompts; the asymmetry gate therefore refuses.

C. Head-to-Head Comparison

Table II compares the gated projection with ungated projection and three common baselines on GPT-2 h.11. This comparison is not presented as proof that the gated method is the strongest edit at every operating point. INLP, for example, achieves a higher raw toxicity reduction in this run. The reliability claim is different: the gated method exposes an

TABLE II

GPT-2 H.11 HEAD-TO-HEAD COMPARISON. TOXICITY VALUES ARE FIVE-SEED MEANS; p IS PAIRED BOOTSTRAP AGAINST BASELINE. THE COMPARISON IS INTERPRETED AS A RELIABILITY AUDIT, NOT AS A FULLY CONTROLLED CLAIM OF UNIVERSAL PERFORMANCE SUPERIORITY.

Method	Tox.	Tox. Red.	Δ PPL	G. Skew	p
Baseline	0.0448	—	—	0.684	—
GBP gated ($s = 0.548$)	0.0208	+53.6%	+1.2%	0.577	< 0.001
GBP ungated ($s = 0.8$)	0.0213	+52.4%	+4.0%	0.677	< 0.001
ActAdd ($\alpha = 3$)	0.0429	+4.1%	−0.2%	0.723	0.370
INLP (8 iter.)	0.0174	+61.2%	+2.6%	0.818	< 0.001
LEACE	0.0369	+17.6%	+0.3%	0.664	0.010

TABLE III

GENDER ABLATION ON GPT-2 H.10. PRONOUN SKEW IS MALE/(MALE+FEMALE); TARGET PARITY IS 0.5. REFUSAL PRESERVES THE BIASED BASELINE, BUT AVOIDS A MORE SEVERE OVER-CORRECTION.

Configuration	Skew	$ \Delta $	Verdict
Baseline	0.704 ± 0.100	0.204	biased baseline
Coherence-only GBP ($s = 0.8$)	0.017 ± 0.033	0.483	over-corrected
INLP	0.309 ± 0.146	0.191	shifted
Three-factor ($s = 0$)	0.704 ± 0.100	0.204	refused

explicit diagnostic and can refuse, while the baselines act whenever invoked.

The gated projection reduces toxicity by +53.6% while paying $3.3\times$ less perplexity cost than ungated projection (+1.2% versus +4.0%). INLP reduces toxicity more in this setting, but it also produces the largest gender side effect (skew 0.818 versus baseline 0.684) and has no pre-flight mechanism for detecting that side effect. ActAdd at $\alpha = 3$ is not statistically significant. These results support the narrower claim: a method can be effective on the target metric and still need a reliability layer.

D. Gender Abstention and Over-Correction

Table III isolates the gender case. The asymmetry gate refuses because every neutral prompt projects to the same side of the gender direction. Removing this gate recreates the failure identified by the reviewers: the method does not debias; it flips the distribution to the opposite extreme.

This result should not be read as a solution to gender bias. Abstention leaves the biased baseline unchanged. The value of the gate is that it identifies the mean-difference direction as unsuitable for projection and prevents an unsafe edit. A bidirectional or distribution-matching gender intervention may be appropriate, but that is a different operator.

E. Cross-Concept Interference

Editing one concept can distort another. At GPT-2 h.11, the dimension-corrected cosine between toxicity and gender directions is $\kappa^*(\hat{\mathbf{v}}_{\text{tox}}, \hat{\mathbf{v}}_{\text{gender}}) = 2.64$ (raw cosine 0.076), yet toxicity editing shifts neutral-prompt gender behavior. This shows that near-orthogonal concept directions can still interact behaviorally. The present framework is a single-concept gate, not a global safety guarantee. A natural extension is a joint

TABLE IV

MISTRAL-7B TOXICITY CALIBRATION. FIVE REPRESENTATIVE LAYERS ARE SHOWN; ALL 32 LAYERS ABSTAIN FROM PROJECTION. HIGH κ^* MEANS PROJECTION WOULD REMOVE TASK SIGNAL ALONG WITH THE CONCEPT DIRECTION.

Layer	κ	κ^*	d_c	s
h.0	0.9897	79.38	0.74	0.000
h.8	0.9983	80.08	0.73	0.000
h.16	0.9890	79.33	0.75	0.000
h.24	0.9419	75.55	0.87	0.000
h.31	0.5014	40.21	1.86	0.000

gate that audits protected concept directions before editing any target concept.

F. Mistral-7B: Projection Abstention Is Operator-Specific

Table IV reports the toxicity diagnostic on Mistral-7B. All 32 layers abstain from projection; five representative layers are shown.

Cohen’s d is respectable (0.72 to 1.86), so the direction is detectable. The failure is coherence: the direction lies too close to the task subspace for a safe projection. However, abstention is not a statement that toxicity is uneditable on Mistral-7B. A focused ActAdd grid on the same model reaches up to +49.4% toxicity reduction at +9.6% perplexity cost. The diagnostic therefore says “projection is the wrong operator here,” not “give up.”

G. Transfer and Scope Summary

Table V summarizes the reported transfer checks. This table deliberately limits the paper’s evidence to the combinations actually reported in the main text, avoiding unsupported claims of universal or unreported concept transfer.

VI. RELATED WORK

a) Activation steering and editing.: ActAdd [1], CAA [2], ITI [4], RepE [3], and spectral activation editing [16] all manipulate hidden representations at inference time. Their focus is usually the construction and tuning of an intervention; our focus is the reliability decision preceding intervention.

b) Linear concept erasure.: INLP [5] and LEACE [6] remove linearly decodable information without a gate. Our experiments show why this matters: INLP can reduce the target metric while creating the largest cross-concept shift in the comparison table.

c) Knowledge editing and localization.: ROME and MEMIT [17], [18] localize and edit factual associations in transformer MLPs. Their target is factual memory rather than behavioral concepts, but both lines of work face the same reliability question: is the edit localized enough to avoid collateral damage?

d) Probing and interpretability.: Linear probing [19] shows that concepts can be decoded from hidden states. Our question is the next step: whether decodability implies editability. Mechanistic interpretability work on superposition and circuits [20] helps explain why a decodable concept can still be entangled with task information.

TABLE V
REPORTED DIAGNOSTIC TRANSFER CHECKS. “DECISION” REFERS TO PROJECTION UNLESS OTHERWISE STATED. RANGES INDICATE SAMPLED OR FULL-LAYER SWEEPS DESCRIBED IN THE TEXT.

Model	Concept	Reported diagnostic evidence	Decision	Interpretation
GPT-2	Toxicity	$\kappa^* = 5.33$, $d_c = 0.73$, $A < 0.01$, $s = 0.548$	Edit	moderate projection; quality limits scale
GPT-2	Sentiment	$\kappa^* = 0.71$, $d_c = 4.52$, $A < 0.01$, $s = 0.799$	Edit	clean direction; near-ceiling scale
GPT-2	Gender	$\kappa^* = 1.84$, $d_c = 1.72$, $A = 1.00$, $s = 0$	Abstain	one-sided direction; projection over-corrects
DistilGPT-2	Toxicity	$\kappa^* = 4.31$, $s = 0.711$ at h.5; earlier layers gated off	Edit	same thresholds transfer without retuning
Pythia-1.4B	Toxicity	sampled-layer $\kappa^* \in [17.79, 56.18]$	Abstain	projection appears task-entangled in sampled layers
Mistral-7B	Toxicity	all-layer $\kappa^* \in [40.21, 80.08]$, $s = 0$	Abstain	projection unsafe; ActAdd can still help
Mistral-7B	Gender	h.0 gives $A = 0.96$ and $s = 0.160$; other layers abstain	Weaken/abstain	high asymmetry warns against displacement

TABLE VI
FACTOR ABLATION ON GPT-2. EACH “DROP” COLUMN FORCES ONE FACTOR TO 1. THE OPERATIVE FACTOR IS THE FACTOR WHOSE REMOVAL MOST CHANGES s .

Concept	L	s_{full}	drop κ^*	drop d_c	drop A	Oper.
Toxicity	h.11	0.548	0.585	0.748	0.548	quality
Gender	h.10	0.000	0.000	0.000	0.798	asym.
Sentiment	h.4	0.799	0.800	0.799	0.799	none

e) Reject-option and reliability methods.: The idea that refusing to act can be the correct decision is established in reject-option classification [21] and selective prediction [22]. This paper imports that principle into activation-space editing and frames it as a reliability layer for inference-time information modification.

VII. DISCUSSION AND LIMITATIONS

a) What the diagnostic does and does not claim.: The fingerprint (κ^*, d_c, A) explains three distinct failure modes: task collision, noisy direction estimation, and one-sided projection. It does not prove that an approved edit is globally safe, optimal, or free of side effects. The gate is a necessary pre-flight check for projection-like operators, not a complete alignment method.

b) Novelty over GBP-Audit.: The contribution is not merely reusing GBP-Audit. The present work corrects coherence for hidden-state dimension, adds a statistical quality factor, introduces an asymmetry trigger, evaluates on text-generation models and activation-editing baselines, and interprets the output as an operator-specific reliability diagnosis. This changes both the method and the scope of the claim.

c) Factor ablation.: Table VI shows that different factors are operative in different cases. Toxicity is limited primarily by direction quality; gender is rejected by asymmetry; sentiment saturates all factors.

d) False abstention and compute cost.: On GPT-2 toxicity, the gate achieves 100% precision (5/5 approved cells are oracle-good) and 83% recall (5/6 oracle-good cells approved) against a grid-search oracle; the false abstention is h.0 at $s = 0.2$, a borderline low-strength case. The three-factor audit runs in 6.3 seconds across all GPT-2 layers, compared with 185.5 seconds for a comparable grid search.

e) Sensitivity.: The midpoint $\kappa_{\text{mid}}^* = 8$ lies between the passing GPT-2 toxicity layer ($\kappa^* = 5.33$) and the failing layers

($\kappa^* > 25$); decisions are invariant over a broad range around the selected value. This is a practical sensitivity result, not a proof of global threshold optimality.

f) Cross-concept limitation.: The gate is single-concept. It can correctly refuse a gender edit and still fail to predict all gender effects of a toxicity edit. A complete deployment system should add a joint audit over protected and task-critical concepts.

g) Scope limitation.: The largest evaluated model is Mistral-7B. The reported transfer evidence includes GPT-2, DistilGPT-2, Pythia-1.4B, and Mistral-7B, but not 70B-class or instruction-tuned frontier systems. The recommendation to report (κ^*, d_c, A) should therefore be read as a practical diagnostic proposal, not as a universal standard already validated across all architectures.

VIII. CONCLUSION

Activation editing should not be treated as a knob that is always safe to turn. We introduced a reliability diagnostic, (κ^*, d_c, A) , that asks whether a concept direction is separable from task information, statistically reliable, and balanced on neutral text before editing. The resulting scale can be zero, so abstention is a valid output.

On GPT-2, the diagnostic permits toxicity projection while reducing perplexity cost relative to ungated projection. On gender, it refuses a one-sided direction that otherwise over-corrects pronoun generation. On Mistral-7B, it correctly diagnoses projection as unsafe while leaving open the possibility of alternative operators such as ActAdd. These results support a narrower but actionable conclusion: activation-steering papers should report pre-flight reliability diagnostics alongside target-metric changes, because target improvement alone does not establish that an edit was safe to perform.

ACKNOWLEDGMENT

This work was supported partly by the DoD Center of Excellence in AI/ML (CoE-AIML) at Howard University under Contract W911NF-20-2-0277 with the U.S. Army Research Laboratory and NSF Grant number 2240407.

However, any opinions, findings, conclusions, or recommendations expressed in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the funding agencies.

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